Consistency Models

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Abstract

Diffusion models generate very good images, but they need many network evaluations to make a single image, which makes them slow. Consistency models [1] are trained to jump from a noisy image straight to the clean one, so they can generate an image with one evaluation. In this project we reproduce them on a small scale with MNIST. We train two "teacher" models (a diffusion model and a Rectified Flow model), distill each one into a consistency model, and also train consistency models from scratch, without a teacher. We compare everything in quality, diversity and speed, using a distance computed with our own digit classifier. With one evaluation, the distilled models are 50 to 95 times faster than the teachers with 50 steps, but their images are clearly worse: their distance is 4 to 5 times larger than that of the best teacher of their family (2.7 and 3.4 times with four steps). The Rectified Flow teacher gives the better consistency model. Our first training from scratch worked badly, but a second run closer to the original paper, with the training images of distillation, matches the distillation of the diffusion teacher with one step and beats it with more. An improved recipe, with the same images, is better still.

1 Introduction

Diffusion models [2, 3] are one of the best ways to generate images today. They learn to remove noise: to create an image, we start from pure noise and clean it little by little, running the neural network at every step. This is slow. The original model [2] uses 1000 steps per image, and later methods based on the *probability flow ODE* [3] reduce this to tens of steps, which is still a lot.

Consistency Models [1] were proposed to solve this. A consistency model takes a noisy image from any point of the denoising path and jumps directly to the clean image at the end of the path, so a single network evaluation is enough. If there is more time, we can use several steps (adding some noise back and cleaning again) to get better images. They can be trained in two ways: *consistency distillation* (CD), where a pre-trained diffusion model acts as a *teacher*, and *consistency training* (CT), where no teacher is used.

In this project we reproduce the method on a small scale with MNIST [4], a dataset light enough to train several models on a CPU, and we measure what is gained and what is lost when fewer steps are used. Besides the initial proposal, we also trained a Rectified Flow [5] teacher and distilled it too, to see how the choice of teacher affects the result.

2 Objectives

The main objectives set in the project proposal are:

1. Train a diffusion model to serve as the teacher of the consistency model.

2. Distill the knowledge learned by the teacher into a faster consistency model.
3. Compare the consistency model with the diffusion model in terms of sampling speed and sample quality.
4. Analyze the results: the speed-up, the quality and diversity of the samples with more or fewer steps, and the limitations we found.

Once these objectives were met, the proposal considered three extensions, in this order: consistency training and its comparison with the previous models; adapting the architecture to color images with CIFAR-10 [6]; and a class-conditional consistency model. Their status is given in Section 6.

3 Related work and state of the art

Diffusion models. Diffusion models [7, 2, 8] were put in a common framework by Song et al. [3] using stochastic differential equations; each one has a deterministic version, the probability flow ODE. Karras et al. [9] studied the design choices (noise schedule, network inputs, ODE solver), and their conclusions are the basis of consistency models and of our code.

Fast sampling. Some methods speed up sampling without extra training, using better ODE solvers such as DDIM [10], the Heun method of EDM [9] or DPM-Solver [11], which gets good quality in around 10 steps. Progressive distillation [12] halves the number of steps in each round, down to 4, but needs several training rounds. Flow matching [13] and Rectified Flow [5] learn paths that are almost straight, which can be followed in fewer steps, and *reflow* straightens them further.

Consistency models. Consistency models [1] learn to jump from any point of the path to its origin, allow one-step generation, and are trained in a single stage, by distillation (CD) or from scratch (CT). Song and Dhariwal [14] improved CT a lot with several changes. Later works apply the idea to text-to-image models [15], to jumps between any two points of the path [16] and to continuous time with 1.5 B parameters [17]. All of them use large GPU budgets; ours is a small reproduction on a CPU that also compares a diffusion and a Rectified Flow teacher.

4 Methodology

We train two teachers on MNIST, a diffusion model of the Variance Exploding type (VE) and a Rectified Flow (RF), and distill each of them into a consistency model. We also train consistency models from scratch, without a teacher: two runs of the original method (CT v1 and its evolution, CT v2) and two with an improved recipe (iCT v1 and v2). All models use the same network, so differences between them come from how they are trained and not from their size. The code is written in PyTorch and is in the project repository.

4.1 Data and architecture

We use the 60 000 MNIST training images (28×28, grayscale, values scaled to $[-1, 1]$, no data augmentation). The 10 000 test images are only used for evaluation (Section 4.6).

All models use the same small U-Net [18] with 1 115 425 parameters (details in Appendix A.1). Only the way its output is used changes: the score network divides it by the noise level $\sigma(t)$, the velocity network uses it as it is, and the consistency models wrap it as explained in Section 4.3.

4.2 Teacher models

In both teachers the clean image is at time $t = 0$ and pure noise at $t = T = 1$, and the training times are drawn from $[\varepsilon, 1]$ with $\varepsilon = 10^{-5}$. Using the same convention lets us train and sample the consistency models with the same code for both teachers.

VE diffusion. The forward process only adds Gaussian noise, $x_t = x_0 + \sigma(t)\, z$ with $z \sim \mathcal{N}(0, I)$ and

$$\sigma(t) = \sqrt{\frac{\sigma^{2t} - 1}{2 \ln \sigma}}, \qquad \sigma = 25, \tag{1}$$

which gives a noise standard deviation of about 9.85 at $t = 1$. The network s_θ (the score) is trained to predict the noise that was added [3], minimizing $\mathbb{E}\, \|\sigma(t)\, s_\theta(x_t, t) + z\|^2$ with t uniform in $[\varepsilon, 1]$. We use two VE teachers. The better one is trained for 60 epochs with Adam, batch size 64 and a learning rate of 10^{-3} with cosine decay to zero, and we keep a smoothed copy of the weights (an exponential moving average, EMA, with rate 0.999). The other one, the *raw* teacher, was trained first and is the one CM-VE was distilled from; it uses 40 epochs, a constant learning rate and no EMA. To generate images we solve the probability flow ODE

$$\frac{dx}{dt} = -\frac{1}{2}\, g(t)^2\, s_\theta(x, t), \qquad g(t) = \sigma^t, \tag{2}$$

from $t = 1$ to $t = \varepsilon$ with Heun's method on the time grid of Karras et al. [9] ($\rho = 7$), spaced in σ and not in t. Each Heun step uses two network evaluations.

Rectified Flow. Rectified Flow [5] joins each image to a noise sample $x_1 \sim \mathcal{N}(0, I)$ with a straight line, $x_t = (1 - t)\, x_0 + t\, x_1$, and trains a velocity network v_θ to predict the (constant) velocity of that line:

$$\mathcal{L}_{\text{RF}} = \mathbb{E}\, \big\| v_\theta(x_t, t) - (x_1 - x_0) \big\|^2, \qquad t \sim \mathcal{U}[\varepsilon, 1]. \tag{3}$$

It is trained for 40 epochs with Adam, batch size 64 and a constant learning rate of 10^{-3}, also with an EMA of the weights (rate 0.999). To generate images we integrate $dx/dt = v_\theta(x, t)$ from $t = 1$ to $t = 0$ with equal Euler steps, one network evaluation each. Heun steps and the cosine decay used for the VE teacher did not improve the samples. The two teachers were tuned separately and with different budgets (60 epochs for VE, 40 for RF), which must be kept in mind when comparing them.

The learned paths are not exactly straight, because the lines of different image–noise pairs cross and the network only learns their average. To straighten them we do one round of *reflow* [5]: we map 60 000 noise samples to images with the first model (1-RF, EMA weights, 100 Euler steps) and retrain on these fixed pairs, whose lines cross much less. The resulting model, the 2-RF, starts from the 1-RF weights and is trained for 10 epochs with a learning rate of $2 \cdot 10^{-4}$.

4.3 Consistency models

Parameterization. A consistency model $f_\theta(x_t, t)$ must give the same clean image for every point of a path, and it must return the input unchanged at $t = \varepsilon$, that is, $f_\theta(x, \varepsilon) = x$. As in the original paper, we guarantee this by construction:

$$f_\theta(x, t) = c_{\text{skip}}(t)\, x + c_{\text{out}}(t)\, F_\theta\big(c_{\text{in}}(t)\, x,\ t\big), \tag{4}$$

where F_θ is the U-Net of Section 4.1. At $t = \varepsilon$, $c_{\text{skip}} = 1$ and $c_{\text{out}} = 0$, so the output is exactly x (coefficients in Appendix A.2).

Consistency distillation. We follow Algorithm 2 of Song et al. [1]. We split $[\varepsilon, T]$ into $N = 150$ times with the Karras spacing (in σ for VE and directly in t for RF). At each iteration, for each image x_0 of the batch, we pick two neighbouring times $t_n < t_{n+1}$, add noise to the image up to $x_{t_{n+1}}$ with the forward process of the teacher, and use the teacher to take one ODE step back to $\hat{x}_{t_n}$ (Heun for VE, Euler for RF). Both points are, approximately, on the same path, so the loss asks the model to give the same output at both:

$$\mathcal{L}_{\text{CD}} = d\big(f_\theta(x_{t_{n+1}}, t_{n+1}),\ f_{\theta^-}(\hat{x}_{t_n}, t_n)\big). \tag{5}$$

The target network f_{θ^-} is an EMA of f_θ (rate $\mu = 0.95$) and receives no gradients. As the distance d we use the pseudo-Huber distance $d(a, b) = \sqrt{\|a - b\|^2 + c^2} - c$ with $c = 0.00054\sqrt{784} \approx 0.015$ [14]. We train for 40 000 iterations with Adam, a constant learning rate of $4 \cdot 10^{-4}$ and batch size 64. We also keep a slower EMA of the weights (rate 0.9999), which is the one saved and used for sampling.

The value of N is our main change with respect to the paper, which uses $N = 18$. During training the network only sees these N times. With $N = 18$ our small U-Net did not work at the times in

between: one-step samples (always at $t = T$) looked fine, but multistep sampling, which adds noise at other times, gave unrecognizable images. With $N = 150$ the problem disappeared.

We distill three models: CM-RF from the EMA weights of the 1-RF, CM-RF (reflow) from the EMA weights of the 2-RF, and CM-VE from the raw VE teacher. *Note that CM-VE was distilled from the raw VE teacher (40 epochs, constant learning rate, no EMA) and not from the better EMA one*; Section 5 reports both teachers and compares CM-VE with the one it was distilled from (recipes in Appendix A.3).

Consistency training. CT (Algorithm 3 of the original paper) trains the model without a teacher, and we only implemented it for the VE process. Instead of the teacher's ODE step, both points of the pair are built with the same noise: $x_{t_{n+1}} = x_0 + \sigma(t_{n+1})\, z$ and $x_{t_n} = x_0 + \sigma(t_n)\, z$, which lie (in expectation) on the same path. The loss and the distance are those of CD, but N and μ follow the schedule of the paper: both grow during training, up to $N = 151$ (formulas in Appendix A.4). We use Adam, batch size 64 and a learning rate of $4 \cdot 10^{-4}$. CT v1 trains for 20 000 iterations with cosine decay and $\mu_0 = 0.95$, and samples with the target network. CT v2 moves closer to the paper's CIFAR-10 setup and to the budget of CD: 40 000 iterations, constant learning rate, $\mu_0 = 0.9$, and a separate EMA of the weights for sampling (rate 0.9995).

Improved consistency training. Because CT v1 was far behind CD, we also trained a model with the main ideas of Song and Dhariwal [14] (formulas in Appendix A.4): no EMA target (the target is the same network, without gradients), a number of times N that doubles in equal stages, noise levels drawn from a lognormal distribution that favours intermediate values, and a loss weighted by $1/(\sigma_{n+1} - \sigma_n)$. Our main run, iCT v2, has the budget of CD and CT v2 (2.6 million training images): 40 000 iterations, constant learning rate, N from 11 to 161 and a sampling EMA of rate 0.9995. An earlier run, iCT v1, used two phases (Appendix A.3): 20 000 iterations with N up to 161 (stage 1), and 32 289 more with batch size 128 and N up to 641, for about 5.4 million images.

4.4 Sampling from consistency models

To generate an image in one step we draw $x_T \sim \mathcal{N}(0, \sigma(T)^2 I)$, with $\sigma(T) \approx 9.85$ for VE and $\sigma(T) = 1$ for RF, and compute $f_\theta(x_T, T)$: one network evaluation. With more steps we use Algorithm 1 of the original paper: for some times $\tau_1 > \cdots > \tau_{n-1}$ we add noise back to the current prediction (clamped to $[-1, 1]$) and clean it again in one jump, so n steps cost n evaluations. The noise is added with the forward process of the teacher, so that x_τ looks like the inputs seen in training (Appendix A.2). The paper chooses the times τ_i with a search that minimizes FID; we did not do this. For VE we use the interior points of a Karras grid in σ with $n + 1$ points ($\tau = 0.16$ for two steps and $\tau = 0.60, 0.16, 0.004$ for four). For RF this rule adds almost no noise (it gives $\tau = 0.027$ for two steps), so we space the times uniformly from $\tau_{\max} = 0.5$ down to ε ($\tau = 0.5, 0.33, 0.17$ for four steps). We chose $\tau_{\max}$ by looking at samples for values between 0.1 and 0.9 (Appendix A.6).

4.5 Differences from the original paper and limitations

Besides the much smaller scale (1.1 M parameters, 40 000 iterations), our work differs from Song et al. [1] in these points: the diffusion teacher is the VE SDE [3] and not an EDM model, and the Rectified Flow teacher is an addition; we use $N = 150$ instead of 18, the pseudo-Huber distance instead of LPIPS or ℓ_2, and re-noising times that are not optimized for FID; the EDM coefficients, designed for $x_t = x_0 + \sigma(t)\, z$, are reused for RF (where the image is also scaled by $1 - t$) so that both teachers share one consistency model, which is reasonable but not optimal; and CT and iCT are only implemented for VE (iCT uses four of the ideas of [14], not dropout or the Fourier embedding scale, and reaches $N = 161$, or 641 in the earlier run, instead of 1280). Instead of FID we use a Fréchet distance on the features of our own classifier (Section 4.6).

4.6 Evaluation protocol

Reference classifier. Inception-based metrics such as FID are designed for natural images, so we trained a small CNN on MNIST (two 3×3 convolutional blocks with 32 and 64 channels, a dense layer of 128 units and 10 outputs; 421 642 parameters; 10 epochs of Adam with learning rate 10^{-3} and batch size 128). It reaches 99.00% accuracy on the test set.

Quality and diversity. Our main quality metric is FD-MNIST, a Fréchet distance computed like FID [19] but using the features of our classifier (its 128-dimensional penultimate layer) instead of Inception, and 10 000 training images as the real reference. Lower is better. It depends on the classifier we chose, so it is not comparable with published FIDs, and we use it together with a visual inspection of the samples. As a second metric we report the fraction of *recognizable* samples (the classifier gives some digit a probability of at least 0.9). The classifier is overconfident with images it has never seen and gives such a probability to 61% of pure-noise images, so this metric must be read with care. Diversity is the entropy of the distribution of predicted digits, with a maximum of $\ln 10 \approx 2.30$ when all digits are equally frequent. Since FD depends on the number of samples, every configuration is evaluated on the same $N_{\text{eval}} = 2000$ samples, generated from the same initial noise within each family (VE or RF). The references are 2000 real test images (the best possible values) and 2000 images of Gaussian noise clamped to $[-1, 1]$.

Timing. We measure the wall-clock time to generate $N \in \{1, 10, 100, 500\}$ images (mean of 3 runs after a warm-up) and report the time per image at $N = 500$, always on the same CPU. The time only depends on the network and on the number of evaluations, so we measured it once per family and number of steps and shared it between models of the same kind. The speed-up is relative to the 50-step teacher of the same family. We also report the number of network evaluations (NFE) per image, which does not depend on the hardware.

5 Results and discussion

Table 1 summarizes quality, diversity and speed; the complete table is in Appendix A.7. Each model was trained only once, so we cannot measure how much the results change between runs. We can measure the noise coming from sampling: evaluating the same models on five other noise draws changes FD by 4 to 12 (Appendix A.8), so differences smaller than about 25 should not be interpreted.

Table 1: Summary of the comparison ($N_{\text{eval}} = 2000$ samples per row). FD-MNIST: lower is better; recognizable and entropy: higher is better. Time per image at $N = 500$ on CPU; speed-up with respect to the 50-step teacher of the same family. CM-VE (CD) was distilled from the raw VE teacher, CM-RF from the 1-RF and CM-RF (reflow) from the 2-RF; CT and iCT use no teacher.

Model	Steps	NFE	FD-MNIST	Recogn. (%)	Entropy (%)	ms/img	Speed-up
Real MNIST (test)	–	–	45.1	97.6	100	–	–
Pure noise	–	–	2947.5	61.0	23	–	–
VE teacher (EMA)	10	20	120.4	79.9	98	13.26	×5.0
VE teacher (EMA)	50	100	125.7	80.8	98	66.43	×1.0
VE teacher (raw)	50	100	205.7	79.4	95	66.43	×1.0
CM-VE (CD)	1	1	456.7	64.2	95	0.70	×95.4
CM-VE (CD)	4	4	323.1	71.7	96	2.68	×24.8
CM-VE (CT v1)	1	1	1922.4	38.5	50	0.70	×95.4
CM-VE (CT v1)	4	4	904.0	50.1	89	2.68	×24.8
CM-VE (CT v2)	1	1	453.5	65.5	98	0.70	×95.4
CM-VE (CT v2)	4	4	267.9	73.2	98	2.68	×24.8
CM-VE (iCT v1)	4	4	189.8	76.6	99	2.68	×24.8
CM-VE (iCT v2)	1	1	417.6	65.4	98	0.70	×95.4
CM-VE (iCT v2)	4	4	159.8	79.5	98	2.68	×24.8
RF teacher (1-RF)	1	1	3791.3	0.0	14	0.66	×50.1
RF teacher (1-RF)	10	10	151.9	78.4	99	6.61	×5.0
RF teacher (1-RF)	50	50	78.5	83.0	100	33.26	×1.0
RF teacher (2-RF)	1	1	533.6	61.4	98	0.66	×50.1
RF teacher (2-RF)	50	50	160.0	76.2	99	33.26	×1.0
CM-RF (CD)	1	1	417.0	66.4	98	0.67	×49.6
CM-RF (CD)	4	4	268.4	75.2	99	2.68	×12.4
CM-RF (reflow)	1	1	394.3	66.6	99	0.67	×49.6
CM-RF (reflow)	4	4	528.4	70.2	99	2.68	×12.4

5.1 Quality versus number of steps

Real test images get an FD-MNIST of 45.1, the best value we can reach with 2000 samples, and pure noise gets 2947.5. The teachers need about 10 steps to make good images (Figure 1). The 1-RF teacher gets 152 with 10 evaluations and 78.5 with 50. The EMA VE teacher is unusable with 5 Heun steps (902) and stays at 120–126 from 10 steps on. The raw VE teacher, the one CM-VE was distilled from, is clearly worse (206 with 50 steps). With one step the 1-RF teacher only outputs the average of the data, a blurry blob (3791), while the straighter 2-RF gets 534.

The distilled models make digits with one evaluation (457 for CM-VE and 417 for CM-RF) and every extra step improves them: 388 and 323 with two and four steps for CM-VE, and 340 and 268 for CM-RF. For a lower cost they are much better than their teachers (CM-VE with 4 evaluations gets 323, while the EMA and raw VE teachers get 902 and 803 with 10), but they do not reach their quality. Compared with the teacher each one was distilled from, CM-VE is 2.2 times worse with one step and 1.6 times with four steps (raw VE teacher, 50 steps), and CM-RF is 5.3 and 3.4 times worse (1-RF). CM-RF is better in absolute terms only because its teacher is much better (78.5 against 206). The recognizable fraction of the one-step models (64–66%) is close to the 61% of pure noise, so this metric says little about them and we rely on FD and on the images.

Reflow barely changes the one-step model (FD 394 against 417, at the limit of the sampling noise), but it makes the multistep ones *worse* (528 and 528 against 340 and 268). Visually, the reflow samples have thinner strokes (Figure 2). The 2-RF teacher is better than the 1-RF with one step (534 against 3791) but worse with many steps (160 against 78.5 with 50); this does not explain the result: CM-RF (reflow) with one step (394) already beats its teacher with one step (534), and then gets worse with more steps while its teacher gets better. We did not investigate the cause; one possibility is that $\tau_{\max}$ was chosen for CM-RF and not for the reflow model. Distilling from the raw weights of the 1-RF instead of the EMA ones gives similar samples but a slightly worse FD (Appendix A.10).

Consistency training without a teacher. CT v1 is far from distillation: FD 1922 with one step (noisy blobs) and 904 with four (Table 3). CT v2 gets 454, 350 and 268 with one, two and four steps. With the same 2.6 million training images, it ties with CM-VE (CD) with one step (457) and beats it with two and four (388 and 323). We changed its settings together, so we cannot say which change matters most. iCT v2, with the same images, gets 418, 269 and 160, the lowest FD of our consistency models with four evaluations. That is better than the raw VE teacher with 50 steps (206), and 1.3 times the best FD of the EMA teacher (120). iCT v1, with 5.4 million images, gets 424, 271 and 190 (692, 425 and 256 after its first stage). Over five noise seeds the two runs only differ clearly with four steps (164 ± 6 against 179 ± 4), so doubling the images did not help, although other settings also differ. CT v2 and iCT are also more diverse than CM-VE (CD) (97–99% of the maximum entropy, against 95–96%). CT v1 saw only 1.3 million images, and CD inherits the limits of a weak teacher.

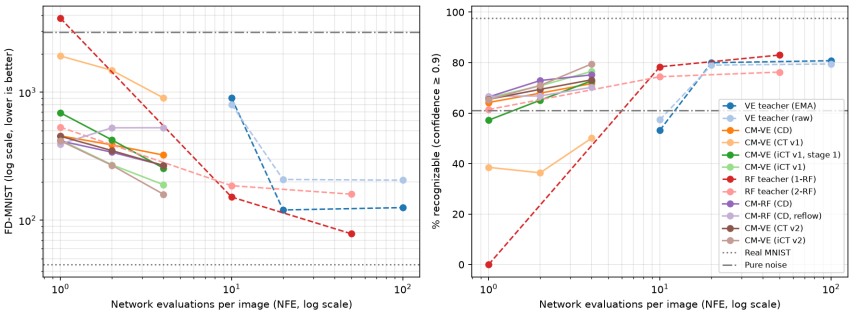


Figure 1: Quality of the samples against the number of network evaluations (NFE): FD-MNIST (left, lower is better) and percentage of recognizable samples (right). Dashed lines are teachers and solid lines consistency models; gray lines are the real-MNIST and pure-noise references.

5.2 Sampling speed

The time per image is dominated by the network evaluations, so the speed-up of each model is almost exactly the ratio of evaluations (Table 1). With one step, CM-VE and CM-RF take 0.70 and 0.67 ms per image, which is $\times 95$ faster than the 50-step VE teacher (66.4 ms, 100 evaluations) and $\times 50$ faster than the 50-step RF teacher (33.3 ms). The total time grows roughly linearly with the number of images N (Figure 3, Appendix A.5); the time per image is almost constant for $N \geq 10$ and larger for $N = 1$ because of fixed overheads, so the speed-ups hold for any batch size. These numbers must be read together with the previous section: a 10-step teacher is already almost as good as a 50-step one, so compared with it (FD 120 for VE and 152 for RF) the one-step speed-up is only $\times 19$ and $\times 10$, with an FD of 457 and 417.

5.3 Diversity

The digits predicted by the classifier (entropy column of Table 1, per-digit shares in Appendix A.7) are close to uniform for the RF teachers with 10 or more steps (99–100% of the maximum entropy) and for all CM-RF models (98–99%), so we see no mode collapse. The VE family is less balanced. The EMA VE teacher already gives only 4% of ones and 16% of twos (real MNIST has 12% and 11%), the raw teacher gives 2% and 17%, and CM-VE (CD) copies the bias of the raw teacher it was distilled from (2% of ones, 95–96% entropy). CT v2 and iCT have no teacher but show a similar lack of ones (4–5%), although they are more balanced overall (97–99%). The low entropy of the one-step 1-RF teacher (14%) does not mean that the generator is bad: its single blurry output is labelled as a 3 in 92% of the cases. In the same way, the one-step CT v1 model makes noise-like images that the classifier labels as 2 or 8 (88% of them), as it does with pure noise.

5.4 Visual inspection

Figure 2 shows three initial noises (ten in Appendix A.9). One-step consistency models make recognizable digits with ragged edges and broken strokes, and more steps clean the strokes and the background (see the 5 of the RF family, third row). They can also change the digit: with four steps CM-VE (CD) breaks the 4 of the second row, CT v2 turns the 5 of the third row into a 9, and iCT v2 turns it into a 4-like shape. The VE consistency models are visibly noisier than the RF ones, the CT v1 samples are noisy blobs even with four steps, CT v2 makes digits comparable to CM-VE (CD), and the four-step iCT samples are the cleanest of the VE family, although many digits are still deformed. With one step, iCT v2 still shows background noise and overlapping strokes.

5.5 Discussion

Teacher choice. CM-RF is better than CM-VE at every number of steps (FD 417, 340 and 268 against 457, 388 and 323) and more diverse, but the comparison is not clean. The RF teacher is much better than the VE one (FD 78.5 against 126 for the EMA teacher and 206 for the raw one that CM-VE was distilled from), the teachers were trained with different budgets, and RF distillation uses Euler steps while VE uses Heun. Compared with its own teacher, CM-VE is the closer copy (2.2 and 1.6 times its FD with one and four steps, against 5.3 and 3.4 for CM-RF). So the advantage of CM-RF comes more from the quality of its teacher than from a better distillation, although we cannot separate it from the type of teacher.

Training without a teacher. CT v1 suggested that CT is far worse than distillation, but CT v2 shows that most of that gap came from its settings and budget: with the images of CD, it reaches or beats CM-VE (CD) without a teacher. The caveat is that this CD is limited by a weak teacher, so we cannot conclude that CT beats distillation in general. iCT v2 improves further with the same images.

Limitations. Besides Section 4.5: every model is a single training run; FD-MNIST uses 2000 samples and our own classifier, so it is not comparable with published FIDs; the timings come from one CPU; and we chose the τ_i by eye. We did not test how much of the gap with the teachers comes from the small network, the distance or the short training.

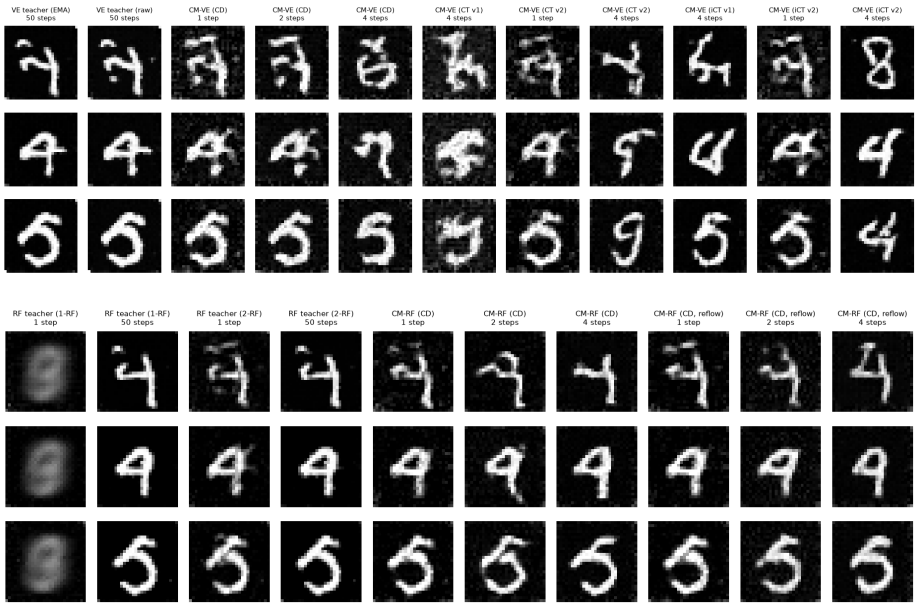


Figure 2: Samples from the VE family (top) and the RF family (bottom). Each column is a configuration and each row starts from the same initial noise.

6 Conclusions and future work

We trained a VE diffusion model and a Rectified Flow model on MNIST, distilled both into consistency models and compared them in quality, diversity and speed, so the four main objectives of the proposal are met. Of the extensions, we did consistency training, in two runs of the original method and with an improved recipe; color images (CIFAR-10) and conditional generation were left out for lack of time. In addition, following a suggestion made during the course, we added the Rectified Flow teacher and a reflow round.

The main results are: (i) one-step consistency models make recognizable digits about 50 to 95 times faster than the 50-step teachers, and more steps improve them smoothly; (ii) with four steps their FD-MNIST is 2.7 (VE) and 3.4 (RF) times worse than that of the best teacher of their family (1.6 for CM-VE compared with the raw teacher it was distilled from), and teachers with 10 to 20 evaluations are already better; (iii) CM-RF is better than CM-VE in absolute terms, but its teacher is also much better and the training budgets differ; (iv) reflow barely changes one-step sampling and, with the re-noising times chosen for CM-RF, hurts multistep sampling; (v) our first CT is far behind CD, but CT v2, with the same training images, ties with CM-VE (CD) with one step and beats it with four (FD 268 against 323), and iCT v2, also with the same images, reaches 160.

Future work includes distilling CM-VE from the EMA teacher, deriving c_{skip} and c_{out} for the RF interpolation, choosing the re-noising times by FD, repeating the trainings with several seeds, and moving to larger networks, a perceptual distance and color images.

References

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A Technical appendices

A.1 U-Net architecture

The encoder has four convolutional levels with 32, 64, 128 and 256 channels (3×3 kernels, no padding, stride 1 in the first level and stride 2 in the other three), and the decoder mirrors it with transposed convolutions; the outputs of the first three encoder levels are concatenated to the inputs of the last three decoder levels. Each level is followed by group normalization (4 groups in the first level, 32 in the others) and the SiLU activation. The time t is encoded with Gaussian random Fourier features (128 fixed random frequencies with scale 30, giving 256 sine/cosine features) followed by a linear layer, and the result goes through SiLU and a dense layer per level whose output is added to the feature maps after every convolution except the last one. Of the 1 115 425 parameters, 128 (the random frequencies) are not trainable. The score network divides the output by $\sigma(t)$; the velocity network and the consistency models use it as it is.

A.2 Consistency model coefficients and re-noising

With $\sigma(t)$ the noise level of the teacher (the VE one, or $\sigma(t) = t$ for RF) and $\Delta(t) = \sigma(t) - \sigma(\varepsilon)$, the coefficients are

$$c_{\text{skip}}(t) = \frac{\sigma_{\text{d}}^2}{\Delta(t)^2 + \sigma_{\text{d}}^2}, \quad c_{\text{out}}(t) = \frac{\sigma_{\text{d}}\,\Delta(t)}{\sqrt{\sigma(t)^2 + \sigma_{\text{d}}^2}}, \quad c_{\text{in}}(t) = \frac{1}{\sqrt{\sigma(t)^2 + \sigma_{\text{d}}^2}}. \tag{6}$$

We use $\sigma_{\text{d}} = 0.5$ as in the original paper, although the standard deviation of MNIST in $[-1, 1]$ is closer to 0.62. The input scaling c_{in} comes from EDM [9]; without it the network would receive inputs with a standard deviation of up to 10 with the VE teacher. In multistep sampling, the re-noising with $z \sim \mathcal{N}(0, I)$ is

$$\text{VE: } x_\tau = x + \sqrt{\sigma(\tau)^2 - \sigma(\varepsilon)^2}\; z, \qquad \text{RF: } x_\tau = (1 - \tau)\, x + \tau\, z. \tag{7}$$

A.3 Training recipes

Table 2: Training recipes. All runs use Adam, seed 42 and a CPU, with batch size 64 unless stated.

Model	Recipe
VE teacher (EMA)	60 epochs, lr 10^{-3} with cosine decay, EMA 0.999
VE teacher (raw, used for CM-VE)	40 epochs, constant lr 10^{-3}, no EMA
RF teacher (1-RF)	40 epochs, constant lr 10^{-3}, EMA 0.999
2-RF (reflow)	60 000 pairs from 100 Euler steps of the 1-RF (EMA); 10 epochs from the 1-RF weights, lr $2 \cdot 10^{-4}$, EMA 0.999
CM-VE, CM-RF, CM-RF (reflow)	CD, 40 000 iterations, constant lr $4 \cdot 10^{-4}$, $N = 150$, $\mu = 0.95$, pseudo-Huber, $\sigma_{\text{d}} = 0.5$, sampling EMA 0.9999
CM-VE (CT v1)	20 000 iterations, lr $4 \cdot 10^{-4}$ with cosine decay, $N(k)$ from 2 to 151, $\mu_0 = 0.95$ ($\mu(k)$ up to 0.9993), pseudo-Huber, sampled with the target network
CM-VE (CT v2)	40 000 iterations, constant lr $4 \cdot 10^{-4}$, $N(k)$ from 2 to 151, $\mu_0 = 0.9$ ($\mu(k)$ up to 0.9986), pseudo-Huber, sampling EMA 0.9995
CM-VE (iCT v1, stage 1)	20 000 iterations, lr $4 \cdot 10^{-4}$ with cosine decay, no EMA target, N from 11 to 161 in 5 stages, lognormal noise levels, weighted pseudo-Huber, sampling EMA 0.999
CM-VE (iCT v1)	stage 1 plus 32 289 iterations from its weights, batch size 128, lr $2 \cdot 10^{-4}$ with cosine decay, N from 161 to 641 in 3 stages, otherwise as stage 1
CM-VE (iCT v2)	40 000 iterations, constant lr $4 \cdot 10^{-4}$, N from 11 to 161 in 5 stages, otherwise as stage 1 except for a sampling EMA of 0.9995

A.4 Consistency training curricula and loss

CT. At iteration k of K, the number of discretization points and the EMA rate of the target network are

$$N(k) = \left\lceil \sqrt{\frac{k}{K}\left((s_1+1)^2 - s_0^2\right) + s_0^2} - 1 \right\rceil + 1, \qquad \mu(k) = \exp\left(\frac{s_0 \ln \mu_0}{N(k)}\right), \tag{8}$$

with $s_0 = 2$, $s_1 = 150$ and $\mu_0 = 0.95$ (v1) or 0.9 (v2), so that training starts with a coarse grid and refines it up to $N(K) = 151$. The grid is the Karras grid in σ with $N(k)$ points.

iCT. The number of points is $N(k) = \min\left(s_0\, 2^{\lfloor k/K' \rfloor},\ s_1\right) + 1$ with $K' = \lfloor K/(\log_2 \lfloor s_1/s_0 \rfloor + 1) \rfloor$, so that N doubles at the start of each of the equal stages: $(s_0, s_1) = (10, 160)$ in stage 1 and iCT v2 and $(160, 640)$ in the second phase of iCT v1. On the same Karras grid $\sigma_1 < \cdots < \sigma_N$, the pair n is drawn with probability

$$p_n \propto \operatorname{erf}\left(\frac{\ln \sigma_{n+1} - P_{\text{mean}}}{\sqrt{2}\, P_{\text{std}}}\right) - \operatorname{erf}\left(\frac{\ln \sigma_n - P_{\text{mean}}}{\sqrt{2}\, P_{\text{std}}}\right), \qquad P_{\text{mean}} = -1.1,\ P_{\text{std}} = 2, \tag{9}$$

and the loss is $\frac{1}{\sigma_{n+1} - \sigma_n}\, d\left(f_\theta(x_{t_{n+1}}, t_{n+1}),\ \operatorname{sg}[f_\theta(x_{t_n}, t_n)]\right)$, with sg the stop-gradient (there is no separate target network) and d the pseudo-Huber distance. As in CT, both points share the same noise z.

A.5 Sampling time

Figure 3 shows that the total sampling time grows roughly linearly with the number of images, so the speed-ups of the main text hold for any batch size.

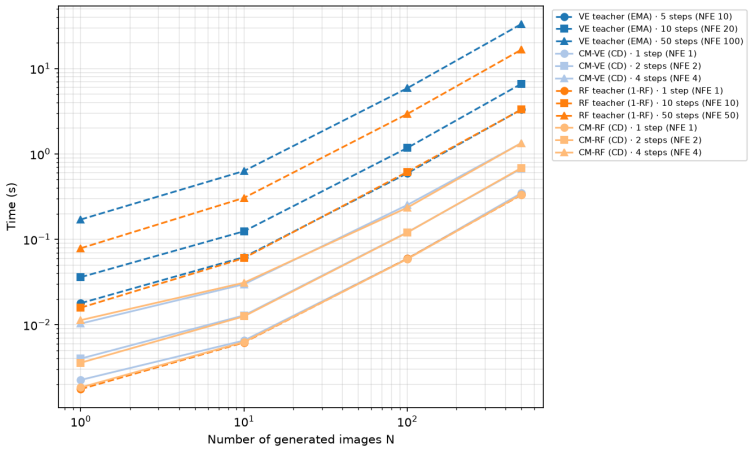


Figure 3: Sampling time versus number of generated samples N (log-log scale). Dashed lines are teachers and solid lines consistency models. Only one model per kind is drawn: models with the same network and number of steps share the same times.

A.6 Re-noising time for RF

Figure 4 shows the sweep used to choose $\tau_{\max}$ for multistep sampling with CM-RF, generated with `system/tests/cm_tau_sweep.py`. Up to $\tau_{\max} = 0.3$ the samples are almost the same as with one step, and from 0.7 on the model regenerates the digit from heavy noise and many digits change or break.

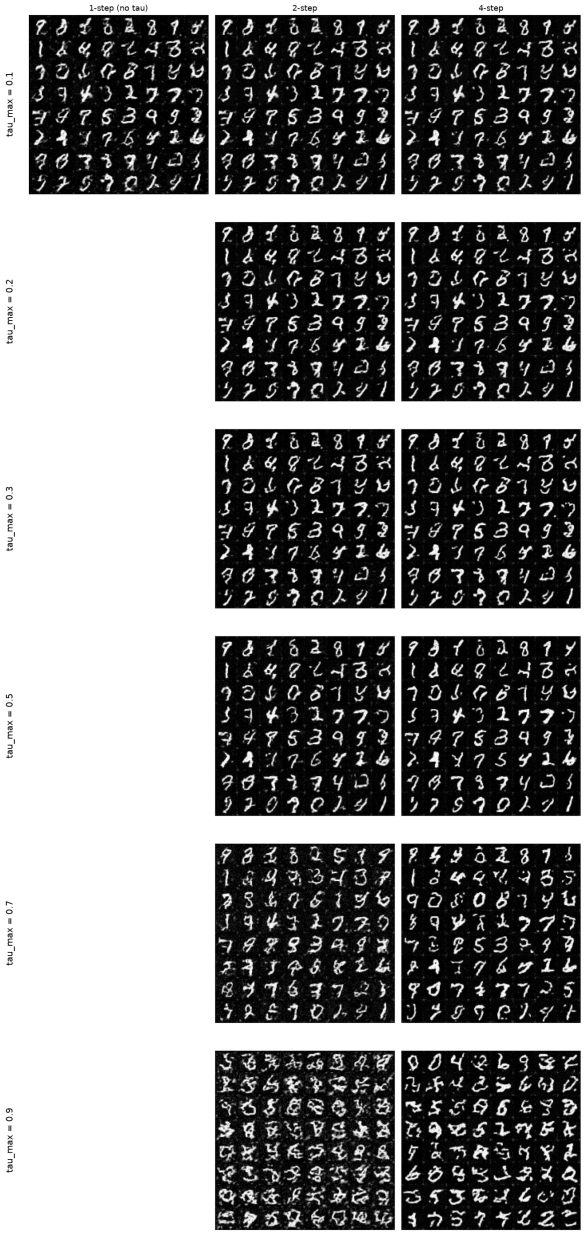


Figure 4: CM-RF samples for different values of $\tau_{\max}$. Each row is a value of $\tau_{\max}$ (0.1, 0.2, 0.3, 0.5, 0.7 and 0.9) and each column a number of steps (2 and 4); the panel at the top left is the one-step result, which does not depend on $\tau_{\max}$. Every panel shows the same 64 initial noises.

A.7 Complete results

Table 3 gives every configuration, including those left out of Table 1. Table 4 and Figure 5 give the share of each digit, which is the detail behind the entropy column.

Table 3: Complete results. Same columns as Table 1. Each consistency model is listed after the teacher it was distilled from (CM-VE from the raw VE teacher, CM-RF from the 1-RF, CM-RF (reflow) from the 2-RF); CT and iCT have no teacher. The times of models with the same network and steps are the same by construction (Section 4.6).

Model	Steps	NFE	FD-MNIST	Recogn. (%)	Entropy (%)	ms/img	Speed-up
Real MNIST (test)	–	–	45.1	97.6	100	–	–
Pure noise	–	–	2947.5	61.0	23	–	–
VE teacher (EMA)	5	10	902.2	53.2	85	6.64	×10.0
VE teacher (EMA)	10	20	120.4	79.9	98	13.26	×5.0
VE teacher (EMA)	50	100	125.7	80.8	98	66.43	×1.0
VE teacher (raw)	5	10	803.1	57.3	84	6.64	×10.0
VE teacher (raw)	10	20	208.5	79.0	95	13.26	×5.0
VE teacher (raw)	50	100	205.7	79.4	95	66.43	×1.0
CM-VE (CD)	1	1	456.7	64.2	95	0.70	×95.4
CM-VE (CD)	2	2	388.4	67.9	96	1.35	×49.3
CM-VE (CD)	4	4	323.1	71.7	96	2.68	×24.8
CM-VE (CT v1)	1	1	1922.4	38.5	50	0.70	×95.4
CM-VE (CT v1)	2	2	1484.1	36.3	76	1.35	×49.3
CM-VE (CT v1)	4	4	904.0	50.1	89	2.68	×24.8
CM-VE (CT v2)	1	1	453.5	65.5	98	0.70	×95.4
CM-VE (CT v2)	2	2	350.4	69.4	98	1.35	×49.3
CM-VE (CT v2)	4	4	267.9	73.2	98	2.68	×24.8
CM-VE (iCT v1, stage 1)	1	1	691.6	57.2	94	0.70	×95.4
CM-VE (iCT v1, stage 1)	2	2	424.7	65.1	94	1.35	×49.3
CM-VE (iCT v1, stage 1)	4	4	255.5	72.8	97	2.68	×24.8
CM-VE (iCT v1)	1	1	423.8	66.1	97	0.70	×95.4
CM-VE (iCT v1)	2	2	271.4	70.8	97	1.35	×49.3
CM-VE (iCT v1)	4	4	189.8	76.6	99	2.68	×24.8
CM-VE (iCT v2)	1	1	417.6	65.4	98	0.70	×95.4
CM-VE (iCT v2)	2	2	268.8	71.0	98	1.35	×49.3
CM-VE (iCT v2)	4	4	159.8	79.5	98	2.68	×24.8
RF teacher (1-RF)	1	1	3791.3	0.0	14	0.66	×50.1
RF teacher (1-RF)	10	10	151.9	78.4	99	6.61	×5.0
RF teacher (1-RF)	50	50	78.5	83.0	100	33.26	×1.0
RF teacher (2-RF)	1	1	533.6	61.4	98	0.66	×50.1
RF teacher (2-RF)	10	10	186.4	74.4	99	6.61	×5.0
RF teacher (2-RF)	50	50	160.0	76.2	99	33.26	×1.0
CM-RF (CD)	1	1	417.0	66.4	98	0.67	×49.6
CM-RF (CD)	2	2	340.4	72.9	99	1.37	×24.2
CM-RF (CD)	4	4	268.4	75.2	99	2.68	×12.4
CM-RF (reflow)	1	1	394.3	66.6	99	0.67	×49.6
CM-RF (reflow)	2	2	527.9	66.8	99	1.37	×24.2
CM-RF (reflow)	4	4	528.4	70.2	99	2.68	×12.4

Table 4: Percentage of samples assigned to each digit by the reference classifier (a perfectly balanced generator gives 10% for each digit).

Model	Steps	0	1	2	3	4	5	6	7	8	9
Real MNIST (test)	–	9	12	11	10	11	9	9	10	9	10
Pure noise	–	0	0	82	2	0	0	0	0	15	0
VE teacher (EMA)	5	4	0	26	17	8	10	3	4	22	6
VE teacher (EMA)	10	11	4	16	13	8	11	9	8	12	9
VE teacher (EMA)	50	12	4	16	12	8	10	10	8	11	9
VE teacher (raw)	5	4	0	26	16	6	11	3	3	23	7
VE teacher (raw)	10	12	2	17	12	6	11	7	6	17	9
VE teacher (raw)	50	13	2	17	11	6	11	8	7	17	9
CM-VE (CD)	1	10	2	17	14	6	13	7	7	16	9
CM-VE (CD)	2	10	2	17	13	7	13	7	8	15	8
CM-VE (CD)	4	13	2	16	12	7	12	8	7	15	8
CM-VE (CT v1)	1	0	0	52	5	2	2	0	0	36	3
CM-VE (CT v1)	2	2	1	23	7	12	7	3	2	38	6
CM-VE (CT v1)	4	6	1	17	8	9	10	9	3	29	8
CM-VE (CT v2)	1	11	4	15	15	7	12	8	9	11	8
CM-VE (CT v2)	2	10	4	16	14	7	11	8	9	11	8
CM-VE (CT v2)	4	12	5	14	14	7	12	8	8	10	10
CM-VE (iCT v1, stage 1)	1	7	2	20	14	9	14	7	8	14	6
CM-VE (iCT v1, stage 1)	2	8	2	19	14	9	14	6	8	15	6
CM-VE (iCT v1, stage 1)	4	10	3	17	15	8	11	9	7	12	9
CM-VE (iCT v1)	1	9	4	16	15	8	13	8	9	11	8
CM-VE (iCT v1)	2	10	4	16	14	7	13	8	9	11	8
CM-VE (iCT v1)	4	11	5	15	13	9	11	9	9	10	9
CM-VE (iCT v2)	1	8	5	15	15	8	13	7	10	12	8
CM-VE (iCT v2)	2	9	5	15	15	8	12	7	10	11	9
CM-VE (iCT v2)	4	10	5	14	14	7	12	9	9	11	9
RF teacher (1-RF)	1	1	1	0	92	0	7	0	0	0	0
RF teacher (1-RF)	10	6	8	10	12	8	11	8	11	11	14
RF teacher (1-RF)	50	8	10	10	11	9	10	9	11	10	12
RF teacher (2-RF)	1	7	8	11	16	8	13	6	12	10	11
RF teacher (2-RF)	10	8	9	11	12	9	11	7	12	10	12
RF teacher (2-RF)	50	8	9	10	12	9	11	8	12	9	12
CM-RF (CD)	1	6	8	11	15	7	12	7	12	11	11
CM-RF (CD)	2	7	10	10	13	8	12	7	12	9	12
CM-RF (CD)	4	7	10	11	13	8	11	7	12	8	12
CM-RF (reflow)	1	6	9	11	12	8	12	6	13	11	12
CM-RF (reflow)	2	7	10	9	13	8	11	6	12	9	13
CM-RF (reflow)	4	8	12	9	12	8	11	7	13	7	14

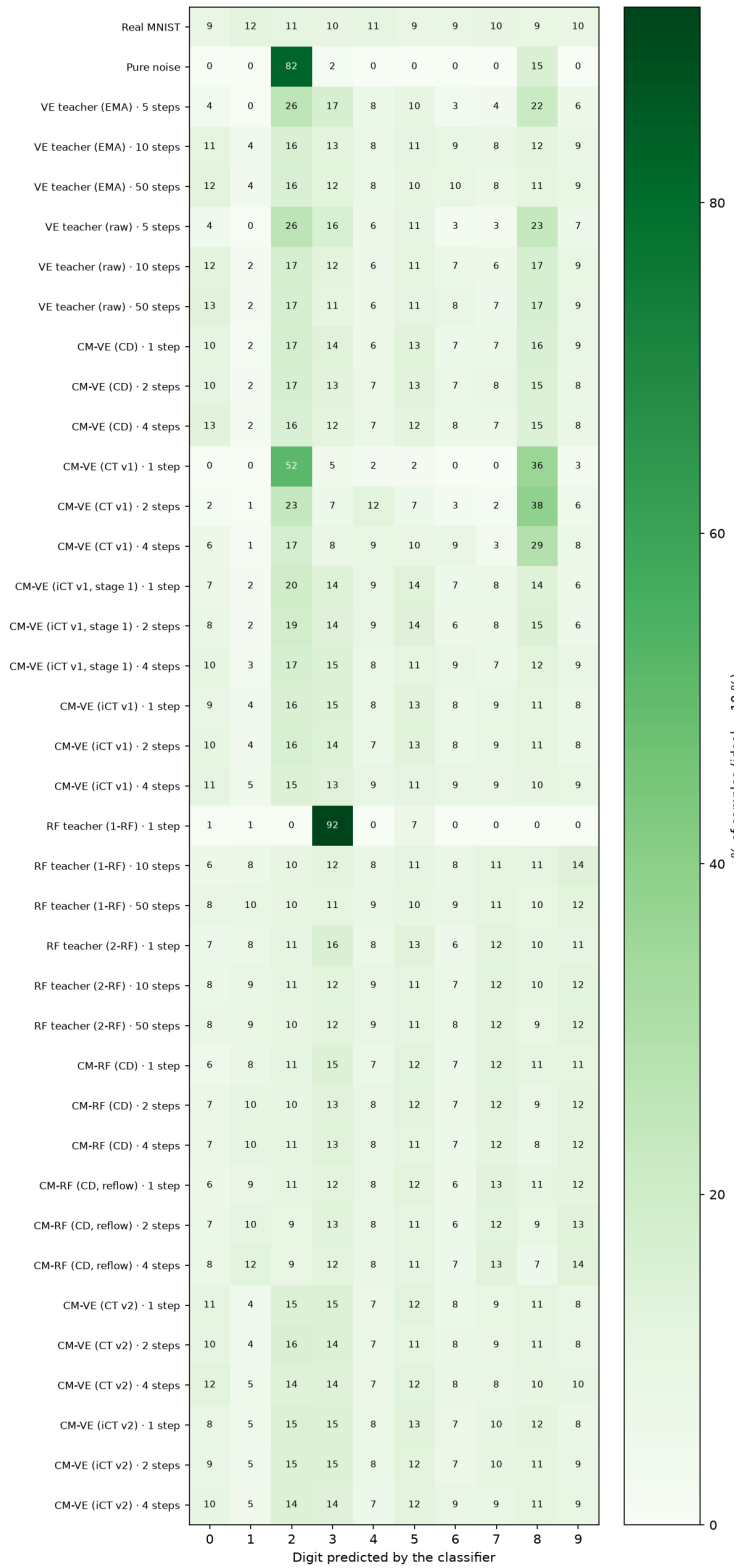


Figure 5: Distribution of the digits predicted by the reference classifier (graphical version of Table 4).

A.8 Sampling variability

Each model was trained once, but the same trained model evaluated on another batch of initial noise gives slightly different metrics. Table 5 repeats the evaluation of the consistency models with five noise seeds (other than the one of the main tables), generated with `system/evaluation/seed_variance.py`. The standard deviations are 4 to 12 for FD-MNIST, at most 1.7 points for the recognizable fraction and at most 1.2 for the entropy, so we do not interpret FD differences below about 25. The values in the main tables use another noise seed (42), which is somewhat favourable: most of them, including the one-step ones, are below these means.

Table 5: Mean and standard deviation over five noise seeds (2000 samples each) for the consistency models.

Model	Steps	FD-MNIST	Recogn. (%)	Entropy (%)
CM-VE (CD)	1	467.8 ± 5.3	64.5 ± 1.0	94.9 ± 0.4
CM-VE (CD)	2	402.5 ± 6.7	66.7 ± 0.9	95.4 ± 0.5
CM-VE (CD)	4	330.0 ± 6.8	71.1 ± 0.8	95.6 ± 0.4
CM-VE (CT v1)	1	1926.9 ± 8.1	38.8 ± 0.6	49.8 ± 1.2
CM-VE (CT v1)	2	1480.6 ± 7.7	37.5 ± 0.7	75.8 ± 0.7
CM-VE (CT v1)	4	903.1 ± 4.8	51.9 ± 0.9	88.4 ± 0.7
CM-VE (CT v2)	1	470.0 ± 7.4	63.5 ± 1.1	97.4 ± 0.4
CM-VE (CT v2)	2	364.0 ± 4.2	67.5 ± 1.0	97.6 ± 0.3
CM-VE (CT v2)	4	268.1 ± 5.7	74.6 ± 0.3	98.2 ± 0.2
CM-VE (iCT v1, stage 1)	1	705.6 ± 7.8	56.2 ± 1.3	94.1 ± 0.7
CM-VE (iCT v1, stage 1)	2	437.7 ± 4.2	64.3 ± 0.8	94.3 ± 0.3
CM-VE (iCT v1, stage 1)	4	261.6 ± 3.9	70.8 ± 0.8	96.5 ± 0.4
CM-VE (iCT v1)	1	438.1 ± 10.7	65.0 ± 1.7	96.9 ± 0.2
CM-VE (iCT v1)	2	284.1 ± 8.6	68.9 ± 0.8	97.3 ± 0.2
CM-VE (iCT v1)	4	178.6 ± 3.9	76.5 ± 0.9	98.3 ± 0.2
CM-VE (iCT v2)	1	429.3 ± 12.0	65.0 ± 1.6	97.5 ± 0.2
CM-VE (iCT v2)	2	276.2 ± 8.1	70.1 ± 1.1	97.6 ± 0.3
CM-VE (iCT v2)	4	164.3 ± 6.2	76.0 ± 1.2	98.6 ± 0.2
CM-RF (CD)	1	435.4 ± 10.7	65.7 ± 0.5	98.2 ± 0.3
CM-RF (CD)	2	353.2 ± 8.1	71.0 ± 0.7	99.0 ± 0.1
CM-RF (CD)	4	272.2 ± 7.0	73.6 ± 0.7	99.0 ± 0.2
CM-RF (reflow)	1	410.6 ± 9.4	66.4 ± 0.8	98.5 ± 0.2
CM-RF (reflow)	2	533.7 ± 12.3	67.8 ± 0.7	98.9 ± 0.1
CM-RF (reflow)	4	547.2 ± 10.9	69.0 ± 0.8	98.9 ± 0.2

A.9 Visual samples with ten initial noises

Figures 6 and 7 extend Figure 2 to all the configurations and ten initial noises. Within a figure, each row starts from the same noise, so the columns can be compared image by image.



Figure 6: Samples from the VE family. Each column is a configuration and each row starts from the same initial noise.

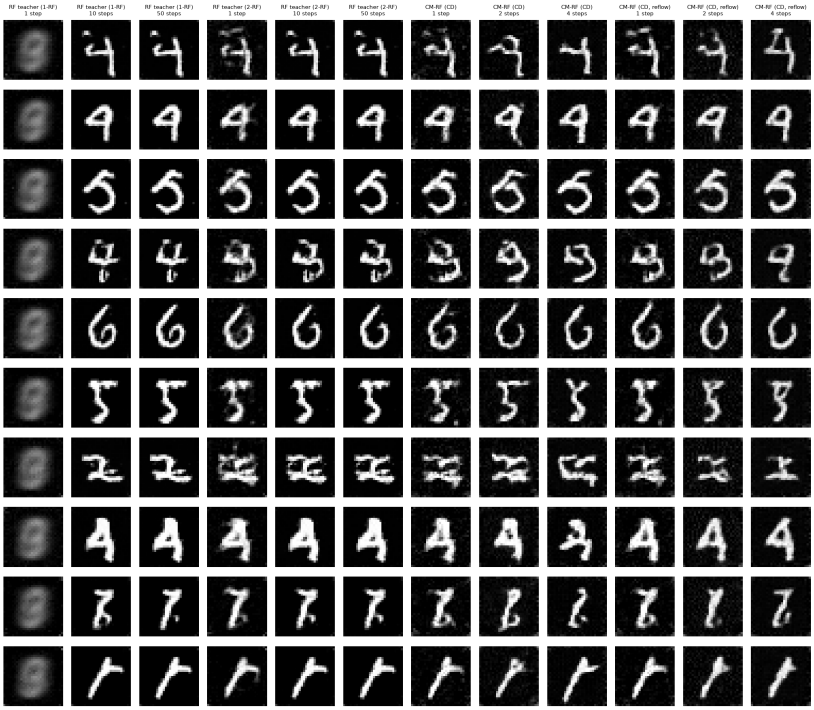


Figure 7: Samples from the RF family. Each column is a configuration and each row starts from the same initial noise.

A.10 EMA and raw teacher weights for CM-RF

Table 6 compares CM-RF distilled from the EMA and from the raw weights of the 1-RF, both evaluated on the same noise, generated with `system/evaluation/cm_rf_teacher_weights.py`. The samples look similar (Figure 8), but the EMA weights give a lower FD with every number of steps. The difference is 16, 29 and 12, so only the two-step one exceeds the threshold of 25 we use (Section 5). The entropy is also about one point higher with EMA. Each model was trained once, so this is not a general result.

Table 6: CM-RF distilled from the EMA or the raw weights of the 1-RF. Mean and standard deviation over five noise seeds (2000 samples each).

Teacher weights	Steps	FD-MNIST	Recogn. (%)	Entropy (%)
EMA	1	435.4 ± 10.7	65.7 ± 0.5	98.2 ± 0.3
EMA	2	354.4 ± 6.1	71.3 ± 0.8	99.0 ± 0.3
EMA	4	276.0 ± 4.8	73.3 ± 1.3	99.0 ± 0.2
Raw	1	451.7 ± 13.0	64.9 ± 1.7	96.9 ± 0.4
Raw	2	382.9 ± 10.2	69.2 ± 1.4	97.7 ± 0.3
Raw	4	287.5 ± 6.1	72.1 ± 0.7	97.9 ± 0.1

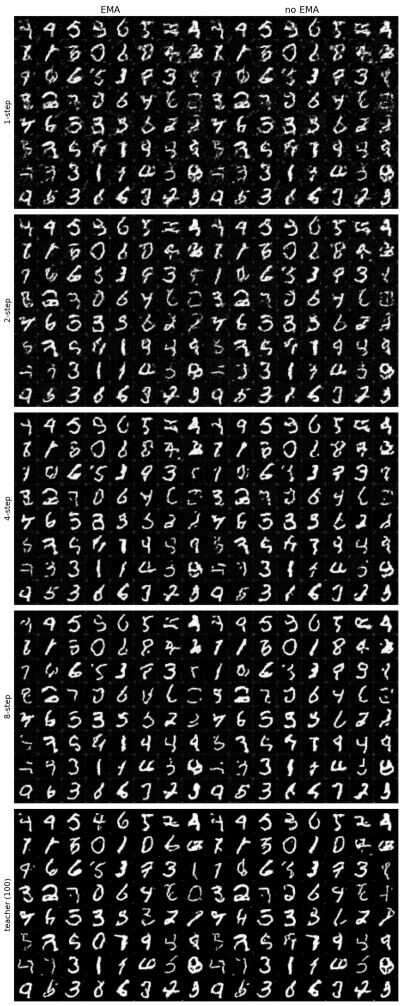


Figure 8: CM-RF distilled from the EMA weights of the 1-RF (left) and from its raw weights (right), with 1, 2, 4 and 8 steps and the 100-step teacher (bottom). All panels start from the same noises.

Author contributions

[PENDIENTE: rellenar por los autores (reparto real del trabajo).]

Declaration regarding the use of AI tools

[PENDIENTE: confirmar por los autores. Borrador: se ha usado un asistente de IA (Claude, de Anthropic) para revisar la memoria contra el código y para redactar y recortar su versión final; los resultados y el código son de los autores.]